METRIC

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SUPERSEDING
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MILITARY SPECIFICATION

NUT, PLAIN, ROUND, BALL AND ROLLER BEARINGS, METRIC, GENERAL SPECIFICATION FOR

Inactive for new design after 1 October 1999

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers nut, plain, round, ball and roller bearings, that is slotted on its periphery, for use in assemblies with ball and roller bearings.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in section 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-H28/21 - Screw-Thread Standards for Federal Services, Section 21, Metric Screw Threads

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-16232 - Phosphate Coating, Heavy, Manganese or Zinc Base (For Ferrous Metals)

DOD-N-70329/1 - Nut, Plain, Round, Ball and Roller Bearings, Narrow Series, Metric

Comments, suggestions, or questions on this document should be addressed to: Defense Supply Center Philadelphia, ATTN: DSCP-NASA, 700 Robbins Avenue, Philadelphia, PA 19111-5096 or emailed to dscpg&ispecomments@dla.mil. Since contact information can change, you may want to verify the currency of this information using the ASSIST Online database at http://assist.daps.dla.mil/

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130 - Identification Marking of US Military Property

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/.or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094)

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NASM 1312 - Fastener Test Methods

(Copies of this document are available from http://www.aia-aerospace.org or the Aerospace Industries Association, 1250 Eye Street NW, Washington DC 20005.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B46.1 - Surface Texture (Surface Roughness, Waviness, and Lay)

(Copies of this document are available from http://www.asme.org or to the American Society of Mechanical Engineers (ASME), Order Department: 22 Law Drive, P.O. Box 2900, Fairfield, NJ 07007-2900.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS STANDARD (ASTM)

ASTM A380 - Standard Practice for Cleaning, Descaling, and Passivation of Stainless Steel Parts, Equipment, and Systems
ASTM B633 - Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel
ASTM E8 - Standard Test Methods for Tension Testing of Metallic Materials
ASTM E1417 - Standard Practice for Liquid Penetrant Examination
ASTM E1444 - Standard Practice for Magnetic Practice Examination

(Copies of this document are available from http://www.astm.org/ ASTM, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

AMERICAN SOCIETY FOR QUALITY

ASQ Z1.4 - Samples Procedures and Tables for Inspection by Attributes

(Copies of this document are available from http://www.asq.org or to the American Society for Quality, 600 Plankinton Avenue, Milwaukee, WI 53203.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE-AMS-STD-66 - Steel: Chemical Composition and Hardenability

(Copies of this document are available from http://www.sae.org or the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

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2.4 Order of precedence. In the event of a conflict between the text of this document and the reference cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Material. Nuts shall be made from the following materials, as specified (see 3.1).

3.2.1 Carbon steel. Nuts shall be made from carbon steel conforming to SAE-AMS-STD-66, steel numbers 1005 to 1035, 1112 to 1115, 1117, 1118, 1211 to 1215. Carbon steel shall have minimum tensile strength of 345 MPa and a minimum yield strength of 172 MPa conforming to ASTM E8.

3.2.2 Corrosion-resistant steel. Nuts shall be made from corrosion-resistant steel conforming to SAE-AMS-STD-66, steel numbers 302, 303, 304 or 316. Corrosion-resistant steel shall have a minimum tensile strength of 579 MPa and a minimum yield of 241 MPa conforming to ASTM E8.

3.2.3 Alloy steel. Nuts shall be made from alloy steel conforming to SAE-AMS-STD-66, steel numbers 4130 to 4142, 8740 or 8742. Alloy steel shall have a minimum tensile strength of 758 MPa and a minimum yield strength of 676 MPa conforming to ASTM E8.

3.3 Design and construction.

3.3.1 Dimensions. Dimensions and tolerances shall conform to the dimensional requirements of the applicable specification sheet and shall apply after protective finishes.

3.3.2 Hardness. Nuts shall have the following hardness:

3.3.2.1 Carbon steel. The hardness of carbon steel shall be 55-92 HRB.

3.3.2.2 Corrosion-resistant steel. Corrosion-resistant steel nuts shall be stress relief annealed and at a hardness of 79-90 HRB.

3.3.2.3 Alloy steel. Alloy steel nuts shall be hardened by quenching in oil from the hardening temperature and tempered to a hardness of 24-32 HRC.

3.3.3 Threads. Threads shall conform to the “M” profile and tolerance class 5H, in accordance with FED-STD-H28/21.

3.3.4 Tensile strength. Nuts shall have the following tensile strength.

3.3.4.1 Carbon steel. Carbon steel nuts shall have a minimum tensile strength and minimum yield strength of 3.2.1.

3.3.4.2 Corrosion-resistant steel. Corrosion-resistant steel nuts shall have a minimum tensile strength and minimum yield strength of 3.2.2.

3.3.4.3 Alloy steel. Alloy steel nuts shall have a minimum tensile strength and minimum yield strength of 3.2.3.

3.4 Surface roughness. Nuts shall have a maximum surface roughness of 3.2 micrometers (μm), determined in accordance with ASME B46.1.

3.5 Protective finish. Nuts shall be protected with one of the following finishes, as specified (see 3.1).

3.5.1 Zinc coating. Zinc coating shall conform to ASTM B633, type II, Fe/Zn 5.

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3.5.2 Phosphate coating. Phosphate coating shall conform to MIL-DTL-16232, type Z, class 2..

3.5.3 Cleaning and descaling. Corrosion-resistant steel shall be cleaned and descaled in accordance with ASTM A380.

3.6 Cracks, seams, inclusions, and flaws. Nuts shall be free from cracks, seams, inclusions, and flaws (see 4.4.5.).

3.7 Sharp edges. Unless otherwise specified on the applicable specification sheet, sharp edges shall be rounded, chamfered or broken, 0.15 to o.40 mm.

3.8 Identification marking. Nuts shall be marked in accordance with MIL-STD-130.

3.9 Workmanship. The workmanship shall be uniform in quality and appearance and free from hanging burrs and slivers, gouges, porosity, objectionable scale or any other defects which may adversely affect the nut's serviceability.

4. VERIFICATION

4.1 Classification of inspections. The inspections specified herein are classified as follows:

a. Material inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 Material inspection. Material inspection shall consist of certification supported by data verifying that the materials used in fabricating nuts are in accordance with the requirements of 3.2 prior to such fabrication.

4.3 Conformance inspection. Conformance inspection shall be in accordance with table I.

TABLE I. Conformance inspection.

Inspection	Requirement Paragraph	Test method Paragraph
Group A		
Examination	1/	4.4.1
Group B		
Hardness	3.3.2	4.4.2
Surface roughness	3.4	4.4.3
Protective finish	3.5	4.4.4
Cracks, seams, inclusions, and flaws	3.6	4.4.5
Group C		
Tensile strength	3.3.4	4.4.6

1/ All requirements not covered by tests.

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4.3.1 Inspection lot. An inspection lot shall consist of all nuts covered by a single specification sheet, produced under essentially the same conditions, and offered for inspection at one time.

4.3.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit it for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.3.3 Sampling for group A inspection. Sampling shall be in accordance with ASQ Z1.4 inspection level II with acceptance on zero and rejection on one for major defects. For minor defects, inspection level I with acceptance on zero and rejection on one.

4.3.3.1 Classification of defects. Defects shall be classified as specified in table II.

TABLE II. Classification of defects.

Category	Defect
Critical	None defined
Major	
101	Runout and parallelism of faces not as Specified.
102	Threads not as specified.
103	Face diameter not as specified.
Minor	
201	Thickness not as specified.
202	Outside diameter not as specified.
203	Protective finish, if applicable, missing.
204	Burrs or sharp edges.
205	Other dimensional characteristics not covered above.
206	Workmanship not as specified.
207	Marking missing, incorrect, insufficient, illegible or not permanent.
208	Packaging not as specified.

4.3.4 Sampling for group B inspection. Sampling shall be in accordance with ASQ Z1.4 inspection level S-2 with acceptance on zero and rejection on one and may be made on sample units which have been subjected to and have passed group A inspection.

4.3.4.1 Disposition of sample units. Sample units which have passed all the group B inspection may be delivered on the contract or purchase order if the lot is accepted and the sample units are still within specified requirements.

4.3.5 Sampling for group C inspection. Sampling shall be in accordance with ASQ Z1.4 inspection level S-2 with acceptance on zero and rejection on one.

4.3.5.1 Disposition of sample units. Sample units which have been subjected to group C inspection shall not be delivered on the contract.

4.3.6 Sampling for protective finish test. Sampling for protective finish test shall be in accordance with the applicable specifications referenced in 3.5.

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4.4 Methods of inspection.

4.4.1 Examination. Sampling nuts shall be thoroughly examined to determine conformance with this specification and applicable specification sheets with respect to all requirements not covered by tests.

4.4.2 Hardness. Nuts shall be tested for hardness in accordance with NASM1312 test method 6.

4.4.3 Surface roughness. Surface roughness shall be tested in accordance with ASME B46.1.

4.4.4 Protective finish test. Test for protective finish shall be in accordance with the applicable specification referenced in 3.5.

4.4.5 Cracks, seams, inclusions, and flaws. The presence of cracks, seams, inclusions, and flaws in nuts shall be determined by one of the following methods of inspection, unless visual inspection disclose discontinuities which would preclude the necessity for these inspection means.

4.4.5.1 Magnetic particle inspection. Magnetic particle inspection shall be performed on noncorrosion-resistant steel nuts in accordance with ASTM E1444.

4.4.5.2 Penetrant inspection. Penetrant inspection shall be performed on corrosion-resistant steel in accordance with ASTM E1417.

4.4.6 Tensile strength. Tensile strength shall be tested in accordance with NASM1312, test method 8 (metric).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. These nuts are primarily intended for use with keyed washers in securing and locking ball and roller bearings in position against shaft shoulders.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification and the applicable specification sheet.
b. Applicable specification sheet part number (see 3.1).
c. Packaging requirements (see 5.1).

6.3 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue..

6.4 Subject term (key word) listing.

Carbon steel, unplated
Carbon steel, zinc coated
CRES
Alloy steel, phosphate coated
Tensile

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Custodians:
Army – AT
Navy – OS
Air Force – 99

Review activities:
Army – CR4
Navy – MC, SH
NSA – NS

Preparing activity:
DLA – IS

(Project 5310 – 2007 – 006)

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